Weekly Problem 9

Ravi Kini

June 1, 2026

Exercise 1

Part (a)

1	$\neg(A \Rightarrow B)$	
2	$\quad A$	
3	$\quad \neg B$	$\neg \Rightarrow$E1, 2
4	$\quad \neg A \vee \neg B$	$\vee$I3
5	$\quad \neg A$	
6	$\quad \neg A \vee \neg B$	$\vee$I5
7	$\neg A \vee \neg B$	TND2–4, 5–6

We then see that $\neg(A \Rightarrow B) \vdash \neg A \vee \neg B$ is a valid argument in connexive logic.

Part (b)

Let $A = i$ and $B = 1$. We see that $\neg A = i$ and $\neg B = 0$, so $\neg A \vee \neg B = i$. However, $A \Rightarrow B = 1$, and so $\neg(A \Rightarrow B) = 0$. Consequently, the argument $\neg A \vee \neg B \vdash \neg(A \Rightarrow B)$ is an invalid argument in connexive logic.